NeuMathe 题库导出

共 43 道题目 | 2025-12-19 17:02

1.

设 $u=f(t)$，而 $t=e^{x}+e^{-y}, f$ 具有二阶连续导数，则 $\frac{\partial^{2} u}{\partial x^{2}}+\frac{\partial^{2} u}{\partial y^{2}}=()$

A $\left(e^{2x}-e^{-2y}\right) f''(t)+\left(e^{x}-e^{-y}\right) f'(t)$

B $\left(e^{2x}+e^{-2y}\right) f''(t)+\left(e^{x}-e^{-y}\right) f'(t)$

C $\left(e^{2x}+e^{-2y}\right) f''(t)+\left(e^{x}+e^{-y}\right) f'(t)$ 【答案】

D $\left(e^{2x}-e^{-2y}\right) f''(t)+\left(e^{x}+e^{-y}\right) f'(t)$

【解析】

$\frac{\partial u}{\partial x}=\frac{\partial u}{\partial t} \frac{\partial t}{\partial x}=f'(t) e^{x}, \frac{\partial u}{\partial y}=\frac{\partial u}{\partial t} \frac{\partial t}{\partial y}=-f'(t) e^{-y}$

$\frac{\partial^{2} u}{\partial x^{2}}=\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)=f''(t) e^{2x}+f'(t) e^{x}$

$\frac{\partial^{2} u}{\partial y^{2}}=\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)=-f''(t)\left(-e^{-y}\right) e^{-y}+f'(t) e^{-y}=f''(t) e^{-2y}+f'(t) e^{-y}$

则 $\frac{\partial^{2} u}{\partial x^{2}}+\frac{\partial^{2} u}{\partial y^{2}}=\left(e^{2x}+e^{-2y}\right) f''(t)+\left(e^{x}+e^{-y}\right) f'(t)$

2.

设 $f(u, v)$ 具有二阶连续偏导数，且满足 $\frac{\partial^{2} f}{\partial u^{2}}+\frac{\partial^{2} f}{\partial v^{2}}=1$，又 $g(x, y)=f\left[x y, \frac{1}{2}\left(x^{2}-y^{2}\right)\right]$，求 $\frac{\partial^{2} g}{\partial x^{2}}+\frac{\partial^{2} g}{\partial y^{2}}=$

A $2y^{2}$

B $x^{2}+y^{2}$ 【答案】

C $2x^{2}$

D $x^{2}-y^{2}$

【解析】

本题是典型的复合函数求偏导问题： $g=f(u, v), \quad u=xy, v=\frac{1}{2}\left(x^{2}-y^{2}\right)$ ，直接利用复合函数求偏导公式即可，注意利用 $\frac{\partial^{2} f}{\partial u \partial v}=\frac{\partial^{2} f}{\partial v \partial u}$.

$$\frac{\partial g}{\partial x}=y \frac{\partial f}{\partial u}+x \frac{\partial f}{\partial v}, \quad \frac{\partial g}{\partial y}=x \frac{\partial f}{\partial u}-y \frac{\partial f}{\partial v}$$

$$\text{故} \frac{\partial^{2} g}{\partial x^{2}}=y^{2} \frac{\partial^{2} f}{\partial u^{2}}+2 x y \frac{\partial^{2} f}{\partial u \partial v}+x^{2} \frac{\partial^{2} f}{\partial v^{2}}+\frac{\partial f}{\partial v}$$

$$\frac{\partial^{2} g}{\partial y^{2}}=x^{2} \frac{\partial^{2} f}{\partial u^{2}}-2 x y \frac{\partial^{2} f}{\partial v \partial u}+y^{2} \frac{\partial^{2} f}{\partial v^{2}}-\frac{\partial f}{\partial v}$$

所以 $\frac{\partial^{2} g}{\partial x^{2}}+\frac{\partial^{2} g}{\partial y^{2}}=\left(x^{2}+y^{2}\right) \frac{\partial^{2} f}{\partial u^{2}}+\left(x^{2}+y^{2}\right) \frac{\partial^{2} f}{\partial v^{2}}$

$$=x^{2}+y^{2}$$

3.

设 $z=x\cdot f\left(x^2+y^2\right)$，求 $\frac{\partial z}{\partial x}=()$ 。

A $f\left(x^2+y^2\right)+2xf'\left(x^2+y^2\right)$

B $f\left(x^2+y^2\right)+2x^2f'\left(x^2+y^2\right)$ 【答案】

C $xf\left(x^2+y^2\right)+2x^2f'\left(x^2+y^2\right)$

D $yf\left(x^2+y^2\right)+2x^2f'\left(x^2+y^2\right)$

【解析】

这里 z 是两个函数的乘积，先用乘法法则再用链式法则

$$\frac{\partial z}{\partial x}=f\left(x^2+y^2\right)+x\cdot\frac{\partial f\left(x^2+y^2\right)}{\partial x}=f\left(x^2+y^2\right)+x\cdot2xf'\left(x^2+y^2\right)$$
$$=f\left(x^2+y^2\right)+2x^2f'\left(x^2+y^2\right)$$

4.

设 $w=e^x\sin y,x=uv,y=u+v$，求 $\frac{\partial w}{\partial v}=()$.

A $ue^x\sin y+e^x\cos y$ 【答案】

B $ue^x\cos y+e^x\sin y$

C $ue^x\sin y+\cos y$

D $e^x\sin y+e^x\cos y$

【解析】

$\frac{\partial w}{\partial v}=\frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial v}+\frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial v}=ue^x\sin y+e^x\cos y$

5.

设 $z=\ln(\sqrt{x}+\sqrt{y})$，则 $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=()$

A $\frac{1}{2}$ 【答案】

B 1

C -1

D 0

【解析】

$$\frac{\partial z}{\partial x}=\frac{1}{2\sqrt{x}(\sqrt{x}+\sqrt{y})},\quad\frac{\partial z}{\partial y}=\frac{1}{2\sqrt{y}(\sqrt{x}+\sqrt{y})}$$
$$x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=\frac{x}{2\sqrt{x}(\sqrt{x}+\sqrt{y})}+\frac{y}{2\sqrt{y}(\sqrt{x}+\sqrt{y})}=\frac{1}{2}$$

6.

设 $w=u^v, u=x+y, v=xy$，求 $\frac{\partial w}{\partial x}=()$.

(A) $xy(x+y)^{xy-1}+x(x+y)^{xy}\ln(x+y)$

(B) $xy(x+y)^{xy-1}+y(x+y)^{y-1}\ln(x+y)$

(C) $xy(x+y)^{xy-1}+y(x+y)^{xy}\ln(x+y)$ 【答案】

(D) $xy(x+y)^{xy-1}+y(x+y)^{xy}\ln(x-y)$

【解析】

用复合函数的求导法则：$\frac{\partial w}{\partial x}=\frac{\partial w}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial w}{\partial v}\cdot\frac{\partial v}{\partial x}=vu^{v-1}\cdot 1+y\cdot u^v\cdot\ln u$. 即有 $xy(x+y)^{xy-1}+y(x+y)^{xy}\ln(x+y)$

7.

设 $z=f\left(\ln x+\frac{1}{y}\right)$，其中函数 $f(u)$ 可微，则 $x\frac{\partial z}{\partial x}+y^2\frac{\partial z}{\partial y}=(\quad)$

(A) 0 【答案】

(B) $2f'$

(C) $-f'$

(D) f'

【解析】

因为 $\frac{\partial z}{\partial x}=f'\cdot\frac{1}{x}, \frac{\partial z}{\partial y}=f'\cdot\left(-\frac{1}{y^2}\right)$，所以 $x\frac{\partial z}{\partial x}+y^2\frac{\partial z}{\partial y}=0$.

8.

设 $z=x^2f\left(xy,\frac{y}{x}\right), f$ 具有二阶连续偏导数，求 $\frac{\partial^2 z}{\partial x\partial y}=()$

(A) $3x^2f_1'-f_2'+x^3yf_{11}''+\frac{y}{x}f_{22}''$

(B) $3x^2f_1'+f_2'+x^3yf_{22}''+\frac{y}{x}f_{11}''$

(C) $3x^2f_1'+f_2'+x^3yf_{11}''-\frac{y}{x}f_{22}''$ 【答案】

(D) $6x^2f_1'+f_2'+x^3yf_{11}''-\frac{y}{x}f_{22}''$

【解析】

$$\frac{\partial z}{\partial y}=x^2\left(xf_1+\frac{1}{x}f_2\right)=x^3f_1'+xf_2'$$

$$\frac{\partial^2 z}{\partial x\partial y}=3x^2f_1'+x^3\left(f_{11}''y-\frac{y}{x^2}f_{12}''\right)+f_2'+x\left(f_{21}''y-\frac{y}{x^2}f_{22}''\right)=3x^2f_1'+f_2'+x^3yf_{11}''-\frac{y}{x}f_{22}''$$

9.

设 $w=e^x\sin y, x=uv, y=u+v$，求 $\frac{\partial w}{\partial u}=()$.

A $ve^x\sin y+\cos y$

B $ve^x\sin y+e^x\cos y$ 【答案】

C $ve^x\sin y-e^x\cos y$

D $ve^x\cos y+e^x\sin y$

【解析】

$\frac{\partial w}{\partial u}=\frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial u}+\frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial u}=ve^x\sin y+e^x\cos y$

10.

设 $f(u)$ 是可导函数，又设 $z=xf(u), u=\frac{y}{x}$，则 $\frac{\partial z}{\partial x}=()$.

A $f\left(\frac{x}{y}\right)+\frac{2y}{3x}f'\left(\frac{x}{y}\right)$

B $f\left(\frac{x}{y}\right)+\frac{x}{y}f'\left(\frac{x}{y}\right)$

C $f'\left(\frac{y}{x}\right)-\frac{y^2}{x}f\left(\frac{y}{x}\right)$

D $f\left(\frac{y}{x}\right)-\frac{y}{x}f'\left(\frac{y}{x}\right)$ 【答案】

【解析】

$\frac{\partial z}{\partial x}=f(u)+xf'(u)\cdot\left(-\frac{y}{x^2}\right)=f\left(\frac{y}{x}\right)-\frac{y}{x}f'\left(\frac{y}{x}\right)$

11.

设 $F(x-y,y-z,z-x)=0$，其中 F 具有连续偏导数，且 $F_2'-F_3'\neq 0$，则 $\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}=()$

A 2

B 1 【答案】

C 0

D -1

【解析】

解答由题意知方程确定函数 $z=z(x,y)$，在题设方程两边取微分
得： $dF(x-y,y-z,z-x)=d0=0$
即有：$F_1'(dx-dy)+F_2'(dy-dz)+F_3'(dz-dx)=0$.
合并得： $(F_1'-F_3')dx+(F_2'-F_1')dy=(F_2'-F_3')dz$
解得：$dz=\frac{F_1'-F_3'}{F_2'-F_3'}dx+\frac{F_2'-F_1'}{F_2'-F_3'}dy$ ，从而 $\frac{\partial z}{\partial x}=\frac{F_1'-F_3'}{F_2'-F_3'}, \frac{\partial z}{\partial y}=\frac{F_2'-F_1'}{F_2'-F_3'}$
于是：$\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}=1$

12.

设 $f(u,v)$ 是二元可微函数， $z=f\left(x^{y},y^{2x}\right)$ ，则 $\frac{\partial z}{\partial x}=$ ().

(A) $yx^{y-1}f_1'+2\ln y\cdot y^{2x}f_2'$ 【答案】

(B) $yx^{y}f_1'+\ln y\cdot y^{2x}f_2'$

(C) $yx^{y-1}f_1'+2xy^{2x-1}f_2'$

(D) $\ln x\cdot x^{y}f_1'+2\ln y\cdot y^{2x}f_2'$

【解析】

由 $z=f\left(x^{y},y^{2x}\right)$ 知 $\frac{\partial z}{\partial x}=yx^{y-1}f_1'+2\ln y\cdot y^{2x}f_2'$

13.

设 $z=xg(x+y)+y\varphi(xy)$，其中 g,φ 具有二阶连续导数，则 $\frac{\partial^2 z}{\partial x\partial y}=$(.

(A) $g'(x+y)+xg''(x+y)+2y\varphi'(xy)+y^2\varphi''(xy)$

(B) $g'(x+y)+xg''(x+y)+2y\varphi'(xy)+xy^2\varphi''(xy).$ 【答案】

(C) $g'(x+y)+g''(x+y)+2y\varphi'(xy)+xy^2\varphi''(xy)$

(D) $g'(x+y)+2y\varphi'(xy)+xy^2\varphi''(xy)$

【解析】

由$z=xg(x+y)+y\varphi(xy)$ 知$\frac{\partial z}{\partial x}=g(x+y)+xg'(x+y)+y^2\varphi'(xy)$

$\frac{\partial^2 z}{\partial x\partial y}=g'(x+y)+xg''(x+y)+2y\varphi'(xy)+xy^2\varphi''(xy)$

14.

设函数 $z=\frac{y}{x}f(xy)$，其中 f 可微，则 $\frac{x}{y}\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}=$(　)

(A) $-2yf'(xy)$

(B) $\frac{2}{x}f(xy)$

(C) $-\frac{2}{x}f(xy)$

(D) $2yf'(xy)$ 【答案】

【解析】

$\frac{x}{y}\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}=\frac{x}{y}\left(-\frac{y}{x^2}f(xy)+\frac{y^2}{x}f'(xy)\right)+\frac{1}{x}f(xy)+yf'(xy)=2yf'(xy).$

15.

设 $z=f(x+y,x-y,xy)$ ，其中 f 具有二阶连续偏导数，则 dz 等于（ ）

A $(f_1'+f_2'+yf_3')dx+(f_1'+f_2'+xf_3')dy$

B $(f_1'+f_2'+xf_3')dx+(f_1'-f_2'+yf_3')dy$

C $(f_1'+f_2'+yf_3')dx+(f_1'-f_2'+xf_3')dy$ 【答案】

D $(f_1'+f_2'+yf_3')dx+(f_1'+f_2'-xf_3')dy$

【解析】

$$\frac{\partial z}{\partial x}=f_1'+f_2'+yf_3',\quad \frac{\partial z}{\partial y}=f_1'-f_2'+xf_3'$$

所以，$dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy=(f_1'+f_2'+yf_3')dx+(f_1'-f_2'+xf_3')dy$

16.

设 $z=xy+u,u=\varphi(x,y)$，则 $\frac{\partial^2 z}{\partial x^2},\frac{\partial^2 z}{\partial x\partial y}$ 分别为 ()

A $1+\varphi_{xx}(x,y),\varphi_{xy}(x,y)$

B $1+\varphi_{xx}(x,y),1+\varphi_{xy}(x,y)$

C $\varphi_{xx}(x,y),1+\varphi_{xy}(x,y)$ 【答案】

D $\varphi_{xx}(x,y),\varphi_{xy}(x,y)$

【解析】

$$\frac{\partial^2 z}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial}{\partial x}\left(y+\frac{\partial u}{\partial x}\right)=\frac{\partial^2 u}{\partial x^2}=\varphi_{xx}(x,y)$$
$$\frac{\partial^2 z}{\partial x\partial y}=\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial}{\partial y}\left(y+\frac{\partial u}{\partial x}\right)=1+\frac{\partial^2 u}{\partial x\partial y}=1+\varphi_{xy}(x,y)$$

17.

设函数 $f(u,v)$ 由关系式 $f[xg(y),y]=x+g(y)$ 确定，其中函数 $g(y)$ 可微，且 $g(y)\neq 0$ ，则 $\frac{\partial^2 f}{\partial u\partial v}=$

A $-\frac{g'(v)}{g^2(v)}$ 【答案】

B $\frac{g'(v)}{g^2(v)}$

C $-\frac{g^2(v)}{g'(v)}$

D $\frac{g^2(v)}{g'(v)}$

【解析】

令 $u=xg(y),v=y$ 可得到 $f(u,v)$ 的表达式，再求偏导数即可。

令 $u=xg(y),v=y$，则 $f(u,v)=\frac{u}{g(v)}+g(v)$，

所以，$\frac{\partial f}{\partial u}=\frac{1}{g(v)},\frac{\partial^2 f}{\partial u\partial v}=-\frac{g'(v)}{g^2(v)}$

18.

设 $z=f\left(x^2+y^2\right)$，其中 f 可微，求 $\frac{\partial z}{\partial x}=()$

A $2yf'\left(x^2+y^2\right)$

B $2xf'\left(x^2+y^2\right)$ 【答案】

C $2xf\left(x^2+y^2\right)$

D $xf'\left(x^2+y^2\right)$

【解析】

这里只有一个中间变量，用求导链式法则只有一项

$$\frac{\partial z}{\partial x}=f'\left(x^2+y^2\right)\cdot\frac{\partial\left(x^2+y^2\right)}{\partial x}=2xf'\left(x^2+y^2\right)$$

19.

已知 $u=f\left(x^2-y^2,e^{x+y}\right)$，其中 f 具有二阶连续偏导数，求 $\frac{\partial^2 u}{\partial x\partial y}=(\quad)$

A $4xyf_{11}'+2(x-y)\mathrm{e}^{x+y}f_{12}'+\mathrm{e}^{2x+2y}f_{22}'+\mathrm{e}^{x+y}f_2'$

B $-4xyf_{11}'+2(x+y)\mathrm{e}^{x+y}f_{12}'+\mathrm{e}^{2x+2y}f_{22}'+\mathrm{e}^{x+y}f_2'$

C $-4xyf_{11}'+2(x-y)\mathrm{e}^{x+y}f_{12}'+\mathrm{e}^{2x+2y}f_{22}'+\mathrm{e}^{x+y}f_2'$ 【答案】

D $4xyf_{11}'+2(x+y)\mathrm{e}^{x+y}f_{12}'+\mathrm{e}^{2x+2y}f_{22}'+\mathrm{e}^{x+y}f_2'$

【解析】

$\frac{\partial u}{\partial x}=2xf_1'+\mathrm{e}^{x+y}f_2',\quad\frac{\partial u}{\partial y}=-2yf_1'+\mathrm{e}^{x+y}f_2'$

$\frac{\partial^2 u}{\partial x\partial y}=2x\left[f_{11}'(-2y)+f_{12}'\mathrm{e}^{x+y}\right]+\mathrm{e}^{x+y}f_2'+\mathrm{e}^{x+y}\left[f_{21}'(-2y)+f_{22}'\mathrm{e}^{x+y}\right]$

$=-4xyf_{11}'+2(x-y)\mathrm{e}^{x+y}f_{12}'+\mathrm{e}^{2x+2y}f_{22}'+\mathrm{e}^{x+y}f_2'$

20.

设 $z=f(x+y,x-y,xy)$，其中 f 具有 2 阶连续偏导数，则 $\frac{\partial^2 z}{\partial x\partial y}$ 等于（）

A $f_3'+f_{11}''-f_{22}''+xyf_{33}''+(x-y)f_{13}''+(x-y)f_{23}''$

B $f_3'+f_{11}''-f_{22}''+xyf_{33}''+(x+y)f_{13}''+(x-y)f_{23}''$ 【答案】

C $f_3'+f_{11}''+xyf_{33}''+(x+y)f_{13}''+(x-y)f_{23}''$

D $f_3'+f_{11}''-f_{22}''+xyf_{33}''+(x+y)f_{13}''+(x+y)f_{23}''$

【解析】

$$\begin{aligned}\frac{\partial z}{\partial x}&=f_1'\cdot 1+f_2'\cdot 1+f_3'\cdot y=f_1'+f_2'+yf_3'\\ \frac{\partial^2 z}{\partial x\partial y}&=\frac{\partial}{\partial y}\left(f_1'+f_2'+yf_3'\right)\\ &=\left(f_{11}''\cdot 1+f_{12}''\cdot(-1)+f_{13}''\cdot x\right)\\ &\quad+\left(f_{21}''\cdot 1+f_{22}''\cdot(-1)+f_{23}''\cdot x\right)\\ &\quad+\left[1\cdot f_3'+y\left(f_{31}''\cdot 1+f_{32}''\cdot(-1)+f_{33}''\cdot x\right)\right]\\ &=f_{11}''-f_{12}''+xf_{13}''+f_{21}''-f_{22}''+xf_{23}''+f_3'+yf_{31}''-yf_{32}''+xyf_{33}''\\ &=f_3'+f_{11}''-f_{22}''+xyf_{33}''+(x+y)f_{13}''+(x-y)f_{23}''\end{aligned}$$

21.

设 $z=e^x+y^2+f(x+y)$，且当 $y=0$ 时，$z=x^3$，则 $\frac{\partial z}{\partial x}=$().

A $e^x+3(x+y)^2+e^{x+y}$

B $e^x+6(x+y)-e^{x+y}$

C $e^x+3(x+y)^2-e^{x+y}$ 【答案】

D $e^x+6(x+y)+e^{x+y}$

【解析】

在等式 $z=e^x+y^2+f(x+y)$ 中令 $y=0$，得 $x^3=e^x+f(x)$，
解得 $f(x)=x^3-e^x$。
则 $f(x+y)=(x+y)^3-e^{x+y}$，代入原式得 $z=e^x+y^2+(x+y)^3-e^{x+y}$。
故 $\frac{\partial z}{\partial x}=e^x+3(x+y)^2-e^{x+y}$。

22.

设 $u=f(x,y,z)$，而 $y=g(x,t),t=h(x,z)$，各函数均有连续偏导数，求 $\frac{\partial u}{\partial x},\frac{\partial u}{\partial z}=()$.

A $f_1'+f_2'(g_1'+g_2'h_1'),\quad f_2'g_2'h_2'-f_3'$

B $f_1'+f_2'g_1'+g_2'h_1',\quad f_2'g_2'h_2'+f_3'$

C $f_1'+f_2'(g_1'+g_2'h_1'),\quad f_2'g_2'h_2'+f_3'$ 【答案】

D $f_2'(g_1'+g_2'h_1'),\quad f_2'g_2'h_2'+f_3'$

【解析】

x 和 z 是自变量，有：

$$\frac{\partial u}{\partial x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\frac{\partial y}{\partial x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\left(\frac{\partial g}{\partial x}+\frac{\partial g}{\partial t}\frac{\partial h}{\partial x}\right)=f_1'+f_2'(g_1'+g_2'h_1')$$

$$\frac{\partial u}{\partial z}=\frac{\partial f}{\partial y}\frac{\partial y}{\partial z}+\frac{\partial f}{\partial z}=\frac{\partial f}{\partial y}\frac{\partial g}{\partial t}\frac{\partial h}{\partial z}+\frac{\partial f}{\partial z}=f_2'g_2'h_2'+f_3'$$

23.

设抽象函数 $z=f\left(x^2-y^2,e^{xy}\right)$，其中 f 偏导数连续，求 $\frac{\partial z}{\partial x}=(\quad)$

A $xf_1'+ye^{xy}f_2'$

B $2xf_1'+xe^{xy}f_2'$

C $2xf_1'+ye^{xy}f_2'$ 【答案】

D $xf_1'-ye^{xy}f_2'$

【解析】

$$\begin{aligned}\frac{\partial z}{\partial x}&=\frac{\partial z}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial x},\quad \text{其中}u=x^2-y^2,\quad v=e^{xy}\\&=f_1'\cdot 2x+f_2'\cdot ye^{xy}=2xf_1'+ye^{xy}f_2'\end{aligned}$$

24.

设 $f(u,v)$ 是二元可微函数，$z=f(x^y,y^x)$，则 $\frac{\partial z}{\partial x}=()$

A $yx^yf_1'-\ln y\cdot y^xf_2'$

B $yx^{y-1}f_1'+xy^{x-1}f_2'$

C $yx^{y-1}f_1'+\ln y\cdot y^xf_2'$ 【答案】

D $\ln x\cdot x^yf_1'+\ln y\cdot y^xf_2'$

【解析】

解答由 $z=f(x^y,y^x)$ 知 $\frac{\partial z}{\partial x}=yx^{y-1}f_1'+\ln y\cdot y^xf_2'$

25.

设函数 $f(u)$ 可微，且 $f'(2)=2$，则 $z=f\left(x^2+y^2\right)$ 在点 $(1,1)$ 处的全微分 $dz|_{(1,1)}=(\quad)$。

A $2dx+4dy$

B $4dx+4dy$ 【答案】

C $4dx+2dy$

D $2dx+2dy$

【解析】

由 $z=f\left(x^2+y^2\right)$，根据全微分公式可得：

$$dz=f'\left(x^2+y^2\right)d(x^2+y^2)=f'\left(x^2+y^2\right)(2xdx+2ydy)$$

将 $x=1,y=1$ 及 $f'(2)=2$ 代入，得：

$$dz|_{(1,1)}=f'(2)(2dx+2dy)=2(2dx+2dy)=4dx+4dy$$

26.

设 $z=u^2+v^2$，而 $u=x+y,v=x-y$，则 $\frac{\partial z}{\partial x},\frac{\partial z}{\partial y}$ 分别为 ().

A $2x,2y$

B $4x,4y$ 【答案】

C x^2,y^2

D x,y

【解析】

$\frac{\partial z}{\partial x}=2u\cdot u'_x+2v\cdot v'_x=2(u+v)=4x,\quad \frac{\partial z}{\partial y}=2u\cdot u'_y+2v\cdot v'_y=2(u-v)=4y$

27.

设 $z=f\left(x^2-y^2,e^{xy}\right)$，其中 f 具有连续二阶偏导数，则 $\frac{\partial z}{\partial x}$ 为（　）。

Ⓐ $2yf_1'+xe^{xy}f_2'$

Ⓑ $yf_1'+xe^{xy}f_2'$

Ⓒ $2xf_1'+ye^{xy}f_2'$ 【答案】

Ⓓ $xf_1'+ye^{xy}f_2'$

【解析】

设 $u=x^2-y^2,v=e^{xy}$，根据复合函数求导法则（链式法则）可得：

$$\frac{\partial z}{\partial x}=f_1'\cdot\frac{\partial u}{\partial x}+f_2'\cdot\frac{\partial v}{\partial x}=f_1'\cdot(2x)+f_2'\cdot(ye^{xy})=2xf_1'+ye^{xy}f_2'$$

28.

设 $z=f(x^2-y^2,e^{xy})$，其中 f 具有连续二阶偏导数，求 $\frac{\partial^2 z}{\partial x\partial y}$ 为（　）。

Ⓐ $-4xyf_{11}''+2(x^2-y^2)e^{xy}f_{12}''+xye^{2xy}f_{22}''+e^{xy}(1+xy)f_2'$ 【答案】

Ⓑ $-4xyf_{11}''+2(x^2-y^2)e^{xy}f_{12}''+xye^{2xy}f_{22}''+e^{xy}(1+y)f_2'$

Ⓒ $4xyf_{11}''+2(x^2-y^2)e^{xy}f_{12}''+xye^{2xy}f_{22}''+e^{xy}(1+xy)f_2'$

Ⓓ $-4xyf_{11}''+2(x^2-y^2)e^{xy}f_{12}''+ye^{2y}f_{22}''+e^{xy}(1+xy)f_2'$

【解析】

令 $u=x^2-y^2,v=e^{xy}$，则

$$\frac{\partial z}{\partial x}=f_1'\cdot\frac{\partial u}{\partial x}+f_2'\cdot\frac{\partial v}{\partial x}=2xf_1'+ye^{xy}f_2'$$

再对 y 求偏导：

$$\begin{aligned}\frac{\partial^2 z}{\partial x\partial y}&=\frac{\partial}{\partial y}(2xf_1')+\frac{\partial}{\partial y}(ye^{xy}f_2')\\&=2x\left[f_{11}''\frac{\partial u}{\partial y}+f_{12}''\frac{\partial v}{\partial y}\right]+\left[(e^{xy}+xye^{xy})f_2'+ye^{xy}\left(f_{21}''\frac{\partial u}{\partial y}+f_{22}''\frac{\partial v}{\partial y}\right)\right]\\&=2x[f_{11}''(-2y)+f_{12}''(xe^{xy})]+e^{xy}(1+xy)f_2'+ye^{xy}[f_{21}''(-2y)+f_{22}''(xe^{xy})]\\&=-4xyf_{11}''+2x^2e^{xy}f_{12}''+e^{xy}(1+xy)f_2'-2y^2e^{xy}f_{21}''+xye^{2xy}f_{22}''\end{aligned}$$

由于 f 具有连续二阶偏导数，故 $f_{12}''=f_{21}''$，合并同类项得：

$$\frac{\partial^2 z}{\partial x\partial y}=-4xyf_{11}''+2(x^2-y^2)e^{xy}f_{12}''+xye^{2xy}f_{22}''+e^{xy}(1+xy)f_2'$$

29.

设 $f(u)$ 具有二阶连续导数，且 $g(x,y)=f\left(\frac{y}{x}\right)+yf\left(\frac{x}{y}\right)$，求 $x^2\frac{\partial^2 g}{\partial x^2}-y^2\frac{\partial^2 g}{\partial y^2}$ （ ）.

【答案】

Ⓐ $\frac{2y}{x}f'\left(\frac{y}{x}\right)$

Ⓑ $\frac{2y}{x}f'\left(\frac{y}{x}\right)+2\frac{y^2}{x^2}f''\left(\frac{y}{x}\right)$

Ⓒ $\frac{y}{x}f'\left(\frac{y}{x}\right)$

Ⓓ $\frac{2y}{x}f'\left(\frac{y}{x}\right)+2\frac{y^2}{x^2}f''\left(\frac{y}{x}\right)$

【解析】

由已知条件 $g(x,y)=f\left(\frac{y}{x}\right)+yf\left(\frac{x}{y}\right)$ 可得:

$$\begin{aligned}
\frac{\partial g}{\partial x}&=f'\left(\frac{y}{x}\right)\cdot\left(-\frac{y}{x^2}\right)+yf'\left(\frac{x}{y}\right)\cdot\frac{1}{y}=-\frac{y}{x^2}f'\left(\frac{y}{x}\right)+f'\left(\frac{x}{y}\right)\\
\frac{\partial^2 g}{\partial x^2}&=-y\left[-2x^{-3}f'\left(\frac{y}{x}\right)+x^{-2}f''\left(\frac{y}{x}\right)\left(-\frac{y}{x^2}\right)\right]+f''\left(\frac{x}{y}\right)\frac{1}{y}\\
&=\frac{2y}{x^3}f'\left(\frac{y}{x}\right)+\frac{y^2}{x^4}f''\left(\frac{y}{x}\right)+\frac{1}{y}f''\left(\frac{x}{y}\right)\\
\frac{\partial g}{\partial y}&=f'\left(\frac{y}{x}\right)\frac{1}{x}+f\left(\frac{x}{y}\right)+yf'\left(\frac{x}{y}\right)\left(-\frac{x}{y^2}\right)=\frac{1}{x}f'\left(\frac{y}{x}\right)+f\left(\frac{x}{y}\right)-\frac{x}{y}f'\left(\frac{x}{y}\right)\\
\frac{\partial^2 g}{\partial y^2}&=\frac{1}{x^2}f''\left(\frac{y}{x}\right)-\frac{x}{y^2}f'\left(\frac{x}{y}\right)-x\left[-y^{-2}f'\left(\frac{x}{y}\right)+y^{-1}f''\left(\frac{x}{y}\right)\left(-\frac{x}{y^2}\right)\right]\\
&=\frac{1}{x^2}f''\left(\frac{y}{x}\right)-\frac{x}{y^2}f'\left(\frac{x}{y}\right)+\frac{x}{y^2}f'\left(\frac{x}{y}\right)+\frac{x^2}{y^3}f''\left(\frac{x}{y}\right)\\
&=\frac{1}{x^2}f''\left(\frac{y}{x}\right)+\frac{x^2}{y^3}f''\left(\frac{x}{y}\right)
\end{aligned}$$

所以

$$\begin{aligned}
x^2\frac{\partial^2 g}{\partial x^2}-y^2\frac{\partial^2 g}{\partial y^2}&=x^2\left[\frac{2y}{x^3}f'\left(\frac{y}{x}\right)+\frac{y^2}{x^4}f''\left(\frac{y}{x}\right)+\frac{1}{y}f''\left(\frac{x}{y}\right)\right]-y^2\left[\frac{1}{x^2}f''\left(\frac{y}{x}\right)+\frac{x^2}{y^3}f''\left(\frac{x}{y}\right)\right]\\
&=\frac{2y}{x}f'\left(\frac{y}{x}\right)+\frac{y^2}{x^2}f''\left(\frac{y}{x}\right)+\frac{x^2}{y}f''\left(\frac{x}{y}\right)-\frac{y^2}{x^2}f''\left(\frac{y}{x}\right)-\frac{x^2}{y}f''\left(\frac{x}{y}\right)\\
&=\frac{2y}{x}f'\left(\frac{y}{x}\right)
\end{aligned}$$

30.

设 $z=f\left(xy,x^2+y^2\right)$，其中 $f(u,v)$ 有二阶连续偏导数，则 $\frac{\partial^2 z}{\partial x\partial y}=()$

A $xyf_{11}+4xyf_{22}$

B $f_1+xyf_{11}+4xyf_{22}$

C $xyf_{11}+2\left(x^2+y^2\right)f_{12}+4xyf_{22}$

D $f_1+xyf_{11}+2\left(x^2+y^2\right)f_{12}+4xyf_{22}$ 【答案】

【解析】

令 $u=xy,v=x^2+y^2$，则

$$\frac{\partial z}{\partial x}=yf_1+2xf_2$$

再对 y 求偏导：

$$\begin{aligned}\frac{\partial^2 z}{\partial x\partial y}&=\frac{\partial}{\partial y}(yf_1+2xf_2)\\&=f_1+y\frac{\partial f_1}{\partial y}+2x\frac{\partial f_2}{\partial y}\\&=f_1+y(f_{11}x+f_{12}2y)+2x(f_{21}x+f_{22}2y)\\&=f_1+xyf_{11}+2y^2f_{12}+2x^2f_{21}+4xyf_{22}\end{aligned}$$

由于 f 有二阶连续偏导数，故 $f_{12}=f_{21}$，整理得：

$$\frac{\partial^2 z}{\partial x\partial y}=f_1+xyf_{11}+2(x^2+y^2)f_{12}+4xyf_{22}$$

31.

求函数 $z=x^2e^y+(x-1)\arctan\left(\frac{y}{x}\right)$ 在点 $(1,0)$ 处的一阶偏导数（ ）

A $(-1,-2)$

B $(-2,-1)$

C $(1,2)$

D $(2,1)$ 【答案】

【解析】

$$z'_x(1,0)=\left[\frac{d}{dx}z(x,0)\right]_{x=1}=\left[\frac{d}{dx}\left(x^2\right)\right]_{x=1}=2$$

$$z'_y(1,0)=\left[\frac{d}{dy}z(1,y)\right]_{y=0}=\left[\frac{d}{dy}\left(e^y\right)\right]_{y=0}=\left[e^y\right]_{y=0}=1$$

32.

设 $f\left(x+y, \frac{y}{x}\right)=x^2-y^2$，$f$ 存在偏导数，则 $\frac{\partial f(x, y)}{\partial x}=$ ().

A $\frac{1-y^2}{x(1+y)^2}$

B $\frac{2y(1+y^2)}{(1-x)^2}$

C $\frac{2x(1-y^2)}{(1+y)^2}$ 【答案】

D $\frac{2y(1-x^2)}{(1+x)^2}$

【解析】

设 $x+y=m, \frac{y}{x}=n$，则 $x=\frac{m}{1+n}, y=\frac{mn}{1+n}$。\n代入原方程得：$f(m, n)=\left(\frac{m}{1+n}\right)^2-\left(\frac{mn}{1+n}\right)^2=\frac{m^2(1-n^2)}{(1+n)^2}$。\n将 m, n 换回 x, y 得：$f(x, y)=\frac{x^2(1-y^2)}{(1+y)^2}$。\n则 $\frac{\partial f(x, y)}{\partial x}=\frac{2x(1-y^2)}{(1+y)^2}$。

33.

设函数 $f(u,v)$ 满足 $f\left(x+y,\frac{y}{x}\right)=x^2-y^2$，则 $\left.\frac{\partial f}{\partial u}\right|_{\substack{u=1\\v=1}}$ 与 $\left.\frac{\partial f}{\partial v}\right|_{\substack{u=1\\v=1}}$ 依次是（　）

A $\frac{1}{2},0$

B $0,\frac{1}{2}$

C $0,-\frac{1}{2}$ 【答案】

D $-\frac{1}{2},0$

【解析】

法一：变量代换法

令 $u=x+y,v=\frac{y}{x}$，则 $x=\frac{u}{v+1},y=\frac{uv}{v+1}$。

代入原方程得：

$$f(u,v)=\left(\frac{u}{v+1}\right)^2-\left(\frac{uv}{v+1}\right)^2=\frac{u^2(1-v^2)}{(v+1)^2}=\frac{u^2(1-v)}{v+1}$$

分别求偏导：

$$\frac{\partial f}{\partial u}=\frac{2u(1-v)}{v+1}$$

$$\frac{\partial f}{\partial v}=u^2\cdot\left[\frac{-(v+1)-(1-v)}{(v+1)^2}\right]=\frac{-2u^2}{(v+1)^2}$$

当 $u=1,v=1$ 时：

$$\left.\frac{\partial f}{\partial u}\right|_{\substack{u=1\\v=1}}=\frac{2(1)(0)}{2}=0$$

$$\left.\frac{\partial f}{\partial v}\right|_{\substack{u=1\\v=1}}=\frac{-2(1)^2}{(1+1)^2}=-\frac{1}{2}$$

法二：复合函数求导法

已知 $f\left(x+y,\frac{y}{x}\right)=x^2-y^2$。

设 $u=x+y,v=\frac{y}{x}$，方程两边分别对 x 和 y 求导。

对 x 求导：

$$\frac{\partial f}{\partial u}\cdot 1+\frac{\partial f}{\partial v}\cdot\left(-\frac{y}{x^2}\right)=2x\quad\cdots(1)$$

对 y 求导：

$$\frac{\partial f}{\partial u}\cdot 1+\frac{\partial f}{\partial v}\cdot\left(\frac{1}{x}\right)=-2y\quad\cdots(2)$$

由 $u=1,v=1$ 可解得 $x=\frac{1}{2},y=\frac{1}{2}$。

代入（1）、（2）式得：

$$\begin{cases}\frac{\partial f}{\partial u}-2\frac{\partial f}{\partial v}=1\\ \frac{\partial f}{\partial u}+2\frac{\partial f}{\partial v}=-1\end{cases}$$

联立解得 $\left.\frac{\partial f}{\partial u}\right|_{\substack{u=1\\v=1}}=0,\left.\frac{\partial f}{\partial v}\right|_{\substack{u=1\\v=1}}=-\frac{1}{2}$。

34.

设 $z=e^u\sin v$，而 $u=xy,v=x+y$，则有关 $\frac{\partial z}{\partial x}$ 和 $\frac{\partial z}{\partial y}$ 的计算正确的是（ ）

(1) $\frac{\partial z}{\partial x}=e^{xy}[y\sin(x+y)+\cos(x+y)]$

(2) $\frac{\partial z}{\partial x}=e^{xy}[y\cos(x+y)+\sin(x+y)]$

(3) $\frac{\partial z}{\partial y}=e^{xy}[x\sin(x+y)+\cos(x+y)]$

(4) $\frac{\partial z}{\partial y}=e^{xy}[x\cos(x+y)+\sin(x+y)]$

A (2)，(3)

B (2)，(4)

C (1)，(3) 【答案】

D (1)，(4)

【解析】

由复合函数求导法则（链式法则）：

$$\begin{aligned}\frac{\partial z}{\partial x}&=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}\\&=e^u\sin v\cdot y+e^u\cos v\cdot 1\\&=e^u(y\sin v+\cos v)\\&=e^{xy}[y\sin(x+y)+\cos(x+y)]\end{aligned}$$

故（1）正确。

$$\begin{aligned}\frac{\partial z}{\partial y}&=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}\\&=e^u\sin v\cdot x+e^u\cos v\cdot 1\\&=e^u(x\sin v+\cos v)\\&=e^{xy}[x\sin(x+y)+\cos(x+y)]\end{aligned}$$

故（3）正确。
综上所述，(1) 和 (3) 的计算是正确的。

35.

设 $z=f(u,x,y),u=xe^y$ ，其中 f 具有二阶连续偏导数，求 $\frac{\partial z}{\partial x}=()$

A $e^yf_1'-f_2'$

B $(e^y-xe^yy_x)f_1'+f_2'+y_xf_3'$

C $e^yf_1'+f_2'$ 【答案】

D $(e^y+xe^yy_x)f_1'+f_2'+y_xf_3'$

【解析】

$\frac{\partial z}{\partial x}=u_xf_1'+f_2'=e^yf_1'+f_2'$

36.

设 $z=x^2f\left(xy,\frac{y}{x}\right),f$ 具有二阶连续偏导数，求 $\frac{\partial z}{\partial y}=()$

A $x^3f_1'-xf_2'$

B $x^3f_2'+xf_1'$

C $x^3f_1'+xf_2'$ 【答案】

D $x^3f_2'-xf_1'$

【解析】

$\frac{\partial z}{\partial y}=x^2\left(xf_1'+\frac{1}{x}f_2'\right)=x^3f_1'+xf_2'$

37.

设 $f(u,v)$ 是二元可微函数，$z=f\left(\frac{y}{x},\frac{x}{y}\right)$ 则 $x\frac{\partial z}{\partial x}-y\frac{\partial z}{\partial y}$ 为（ ）

A $2\frac{y}{x}f_1'\left(\frac{y}{x},\frac{x}{y}\right)-2\frac{x}{y}f_2'\left(\frac{y}{x},\frac{x}{y}\right)$

B $-\frac{y}{x}f_1'\left(\frac{y}{x},\frac{x}{y}\right)+\frac{x}{y}f_2'\left(\frac{y}{x},\frac{x}{y}\right)$

C $-2\frac{y}{x}f_1'\left(\frac{y}{x},\frac{x}{y}\right)+2\frac{x}{y}f_2'\left(\frac{y}{x},\frac{x}{y}\right)$ 【答案】

D $-2\frac{y}{x}f_1'\left(\frac{y}{x},\frac{x}{y}\right)-2\frac{x}{y}f_2'\left(\frac{y}{x},\frac{x}{y}\right)$

【解析】

利用求导公式可得

$$\frac{\partial z}{\partial x}=-\frac{y}{x^2}f_1'+\frac{1}{y}f_2',\quad \frac{\partial z}{\partial y}=\frac{1}{x}f_1'-\frac{x}{y^2}f_2',$$

所以 $x\frac{\partial z}{\partial x}-y\frac{\partial z}{\partial y}=-2\left(f_1'\frac{y}{x}-f_2'\frac{x}{y}\right)$.

38.

设 x,y 是由 $x^2-y^2+2r=0$ 及 $xy-s=0$ 确定的 r,s 的函数，则 $\frac{\partial x}{\partial r}$ 等于（ ）

A $\frac{y}{x^2+y^2}$

B $-\frac{y}{x^2+y^2}$

C $\frac{x}{x^2+y^2}$

D $-\frac{x}{x^2+y^2}$ 【答案】

【解析】

$\begin{cases}2x\mathrm{d}x-2y\mathrm{d}y+2\mathrm{d}r=0\\ x\mathrm{d}y+y\mathrm{d}x-\mathrm{d}s=0\end{cases}$ 可得 $\mathrm{d}x=\frac{y\mathrm{d}s-x\mathrm{d}r}{x^2+y^2}$，故 $\frac{\partial x}{\partial r}=-\frac{x}{x^2+y^2}$

39.

设 $u=x\varphi(x+y)+y\phi(x+y)$，其中函数 φ,ϕ 具有二阶连续导数，$\frac{\partial^2 u}{\partial x^2}-2\frac{\partial^2 u}{\partial x\partial y}+\frac{\partial^2 u}{\partial y^2}=$（　）。

Ⓐ 1

Ⓑ -1

Ⓒ 0 【答案】

Ⓓ 2

【解析】

$$\frac{\partial u}{\partial x}=\varphi(x+y)+x\varphi'(x+y)+y\phi'(x+y),\quad \frac{\partial u}{\partial y}=\phi(x+y)+x\varphi'(x+y)+y\phi'(x+y)$$
$$\frac{\partial^2 u}{\partial x^2}=2\varphi'(x+y)+x\varphi''(x+y)+y\phi''(x+y)$$
$$\frac{\partial^2 u}{\partial y^2}=2\phi'(x+y)+x\varphi''(x+y)+y\phi''(x+y)$$
$$\frac{\partial^2 u}{\partial x\partial y}=\varphi'(x+y)+\phi'(x+y)+x\varphi''(x+y)+y\phi''(x+y)$$

比较上述三式易见：

$$\frac{\partial^2 u}{\partial x^2}-2\frac{\partial^2 u}{\partial x\partial y}+\frac{\partial^2 u}{\partial y^2}=0$$

40.

已知 $u=f\left(x^2-y^2,e^{x+y}\right)$，其中 f 具有二阶连续偏导数，求 $\frac{\partial u}{\partial x}=$（　）。

Ⓐ $2xf_1'+e^{x+y}f_2'$ 【答案】

Ⓑ $2xf_1'-y^2f_1'+e^{x+y}f_2'$

Ⓒ $2xf_1'-y^2f_1'-e^{x+y}f_2'$

Ⓓ $2xf_1'-e^{x+y}f_2'$

【解析】

设 $v=x^2-y^2,w=e^{x+y}$，则 $u=f(v,w)$。
根据多元复合函数求导法则（链式法则）：

$$\frac{\partial u}{\partial x}=\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}+\frac{\partial f}{\partial w}\cdot\frac{\partial w}{\partial x}$$

其中 $\frac{\partial v}{\partial x}=2x$，$\frac{\partial w}{\partial x}=e^{x+y}$，且记 $\frac{\partial f}{\partial v}=f_1',\frac{\partial f}{\partial w}=f_2'$。
代入可得：

$$\frac{\partial u}{\partial x}=f_1'\cdot 2x+f_2'\cdot e^{x+y}=2xf_1'+e^{x+y}f_2'$$

41.

设函数 $f(u)$ 可微，且 $f'(0)=\frac{1}{2}$，则 $z=f\left(4x^2-y^2\right)$ 在点 $(1,2)$ 处的全微分 $\mathrm{d}z|_{(1,2)}=(\quad)$.

(A) $4dx-2dy$ 【答案】

(B) $2dx+4dy$

(C) $2dx-4dy$

(D) $4dx+2dy$

【解析】

方法一：因为 $\frac{\partial z}{\partial x}\Big|_{(1,2)}=f'\left(4x^2-y^2\right)\cdot 8x|_{(1,2)}=4$,

$$\frac{\partial z}{\partial y}\Big|_{(1,2)}=f'\left(4x^2-y^2\right)\cdot(-2y)|_{(1,2)}=-2,$$

所以 $\mathrm{d}z|_{(1,2)}=\left[\frac{\partial z}{\partial x}\Big|_{(1,2)}\mathrm{d}x+\frac{\partial z}{\partial y}\Big|_{(1,2)}\mathrm{d}y\right]=4\mathrm{d}x-2\mathrm{d}y$.

方法二：对 $z=f\left(4x^2-y^2\right)$ 微分得

$$\mathrm{d}z=f'\left(4x^2-y^2\right)\mathrm{d}\left(4x^2-y^2\right)=f'\left(4x^2-y^2\right)(8x\mathrm{d}x-2y\mathrm{d}y),$$

故 $\mathrm{d}z|_{(1,2)}=f'(0)(8\mathrm{d}x-2\mathrm{d}y)=4\mathrm{d}x-2\mathrm{d}y$.

42.

已知函数 $f(u)$ 具有二阶导数，且 $f'(0)=1$，函数 $y=y(x)$ 由方程 $y-xe^{y-1}=1$ 所确定，设 $z=f(\ln y-\sin x)$，求 $\frac{\mathrm{d}z}{\mathrm{d}x}\Big|_{x=0}=(\quad)$.

(A) 2

(B) 0 【答案】

(C) 1

(D) -1

【解析】

令 $u=\ln y-\sin x$，根据复合函数求导法则，$\frac{\mathrm{d}z}{\mathrm{d}x}=f'(u)\cdot\frac{\mathrm{d}u}{\mathrm{d}x}=f'(u)\cdot\left(\frac{1}{y}y'-\cos x\right)$。

由方程 $y-xe^{y-1}=1$，当 $x=0$ 时，解得 $y=1$。

方程两边对 x 求导，得 $y'-\left(e^{y-1}+xe^{y-1}y'\right)=0$。

代入 $x=0,y=1$，得 $y'(0)-e^0=0$，即 $y'(0)=1$。

当 $x=0$ 时，$u=\ln 1-\sin 0=0$。

故 $\frac{\mathrm{d}z}{\mathrm{d}x}\Big|_{x=0}=f'(0)\cdot\left(\frac{1}{1}\cdot 1-\cos 0\right)=1\cdot(1-1)=0$。

43.

设 $z=f\left(xy,x^2+y^2\right)$，其中 f 可微，则 $\frac{\partial z}{\partial x}=(\quad)$。

(A) $\frac{\partial z}{\partial x}=yf_1'+2xf_2'+y^2f_2'$

(B) $\frac{\partial z}{\partial x}=yf_1'+2yf_2'$

(C) $xf_1'+2xf_2'$

(D) $\frac{\partial z}{\partial x}=yf_1'+2xf_2'$ 【答案】

【解析】

令 $u=xy,v=x^2+y^2$，则 $z=f(u,v)$。根据多元复合函数求导法则（链式法则）：

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}$$

因为 $\frac{\partial u}{\partial x}=y$，$\frac{\partial v}{\partial x}=2x$，且记 $f_1'=\frac{\partial f}{\partial u},f_2'=\frac{\partial f}{\partial v}$，
所以 $\frac{\partial z}{\partial x}=yf_1'+2xf_2'$。